

RX RevolutionX

DESIGN & ENGINEERING PORTFOLIO



Ford

Carsales

Bosch

Leap Australia

Zagame

Trinity

Objective 3D

Ansys

WELCOME TO OUR JOURNEY

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INITIAL OBJECTIVES

Before beginning development, we established overarching objectives:

- **Race Performance:** Achieve best track times.
- **Compliance:** Comply with all regulations.
- **Durability:** Endure all races without repair.

FREE BODY DIAGRAM

LAUNCH EFFICIENCY

The energy conserved from initial launch is directly correlated to the track times. We found that the canister only applies thrust for approx 1/3 of track length.

TRACK TESTING

From testing force output of CO2 canisters, we found a 20% error range. Physical testing for times was deemed inconsistent. Instead, we tested individual factors.

MATHEMATICAL RACE MODEL

To evaluate designs efficiently, we created a Python race model including friction, yaw stability, and aerodynamics.

MATH MODEL CODE

REFINE MODE

A refine mode allowed investigating single variables. This produced graphs of race times vs variable changes.

RACE MODEL OUTPUT GRAPHS

KEY FINDINGS

The model allowed us to weight variables. For example, increased wheelbase drag negated stability benefits.

KEY PERFORMANCE INDICATORS

PIE CHART: KPIS

THRUST EFFICIENCY

TIPPING MOMENT

Tipping moment creates load on front wheels. We aim to align center of mass with thrust vector within 5mm vertically.

INERTIA

Inertia is resistance to change in velocity. Rotational inertia is quadratic to radius. 41.65% of performance is dictated by mass.

AERODYNAMIC EFFICIENCY

ADVERSE PRESSURE GRADIENTS

Regions of airflow recirculation creating drag.

DIAGRAM

WHEEL WAKE

Rotating surfaces pull airflow into high pressure regions.

MAGNUS EFFECT

Caused by deflection of air from counter-directional spin.

MAGNUS EFFECT DIAGRAM

SUMMARY OF KEY DESIGN OBJECTIVES

- » **Car Mass:** Close to 50g limit.
- » **Durability:** Withstand 10 races.
- » **Aerodynamics:** Reduce adverse pressure.
- » **Tipping:** CoM within 5mm of thrust.

BEARING RESEARCH

tested various bearing types and lubricants. Used a machine to spin to 12,000 RPM.

BEARING RIG

BURNISHING INVESTIGATION

Hybrid ceramic bearings performed consistently better. 21.53% increase in spin time after burnishing.

GRAPH: BURNISHING REVS VS TIME

LUBRICATION & SEALS

Removal of seals initially better, but contamination risk high.

CONDITION	TRIAL 1	AVG
Seals	119.63	117.17
Removed	114.73	114.97
Dust	33.50	34.34

LUBRICANT	TRIAL 1	TRIAL 2	AVG
Factory	119.63	116.74	117.17
Light Oil	27.58	28.38	27.84
Isopropyl	112.51	104.30	106.28
Clean	126.66	119.38	123.95

SMR73C-2OS #7 LD hybrid ceramic selected.

APPLICATION TO RACE MODEL

Derived bearing friction function applied to model.

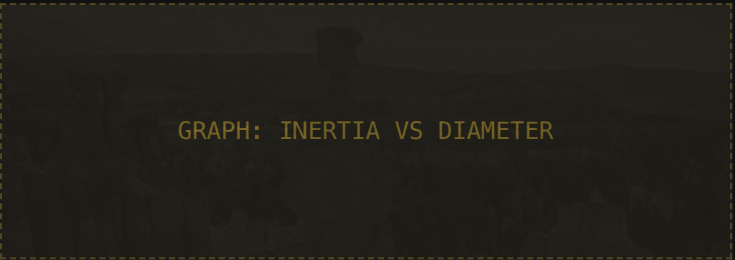
F_f = (95000 / (t + 8.5)) ...

MOMENT OF INERTIA

Wheels must accelerate to 1400 rad/s. Minimizing Mol is key.

WHEEL RADIUS & THICKNESS

Mass distribution determines moment of inertia. We investigated effects using Fusion 360.



YIELD STRENGTH

Limit deformation to elastic region. Peak stress 60% of yield at 50N.

DEFORMATION

Rolling resistance is due to deformation. Stiff wheels are a key goal.

TRACK TESTING

Trade-off between low Mol and stiffness. 1.28g wheels outperformed 0.80g wheels due to stiffness.

TYPE	TIME
Flexible	1.185
Rigid	1.182

SUMMARY OF OBJECTIVES

- Durability (60% Yield)
- Compliance
- Stiffness (<0.25mm)
- Low Inertia

MATERIAL SELECTION

Compared Nylon 12, PEEK, Accura. Selected Ketron PEEK 1000.

MATERIAL	TENSILE	DENSITY
Nylon 12	46	1.01
PEEK	110	1.31

DESIGNS & EVALUATIONS

5 Hole Design



3 Spoke Design



Rib running along inside was added to 3-spoke.

5 SPOKE DESIGN



Stress: 32.42 MPa
Mol: 168.86 gmm
Stress: 57.02 MPa. Mol: 156.85. Best balance.

TOPOLOGICAL OPTIMISATION



Stress: 123. MPa
Failed criteria.
Irregular shape impossible to machine.

FINAL WHEEL SELECTION

Selected 5 spoke design. Met all criteria.

PRIOR DEVELOPMENT & TESTING

Viewed World Finals as continuation. Used ANSYS Fluent. Reproducibility between software confirmed.

Fluent: 0.226NDiscovery: 0.211N

NATIONAL FINALS CAR

Analysis of previous car provided insight. Confirmed accuracy of CFD to physical testing.



Physical: 0.19N. CFD: 0.21N.

DESIGN EVALUATION

Convex-concave nosecone effective. Large fillet reduced effectiveness.

DESIGN PROCESS

Individual development labelled A, B, C, D. Combined later.

PROTOTYPE A



CONCEPT AIM

Features convex-concave features. Redirects air to top of side components.



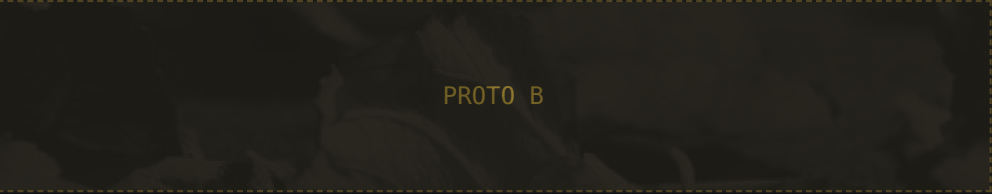
DESIGN EVALUATION

Open channel resulted in additional drag. Front wing support weak.

IMPROVEMENT ACTIONS

Create physical barrier between front wheels and central airflow.

PROTOTYPE B



CONCEPT AIM

Gradually curved streamlined body shape. Unifies components.

DESIGN ANALYSIS

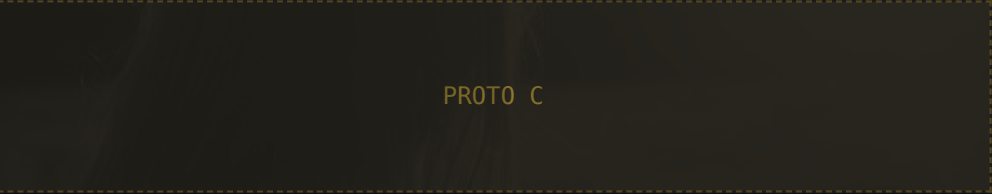
Drag: 0.3088N. Lift: -0.0611N.



DESIGN EVALUATION

Even pressure distribution achieved. Side pods caused rollover.

PROTOTYPE C



CONCEPT AIM

Body shaped to cover side of front wheels. Re-energising airflow.

ANALYSIS

Drag: 0.2931 N. Lift: -0.1443 N.



EVALUATION

Side pod inlet proved ineffective. Increased pressure on nosecone.

IMPROVEMENT ACTIONS

Reduce height of nosecone. Remove inlets.

PROTOTYPE D



IDEA

Composite body shape. Concave centre. Directs outboard.

ANALYSIS

Drag: 0.2868 N. Best so far.



COMBINED PROTOTYPE

Combined features A, B, C, D. Convex-concave nosecone + channels.

ITERATIVE COMPONENT DEVELOPMENT

FRONT WING

Twisted front wing proposed to align with airflow.

SWEPT ENDPLATE

Prevent rollover between pressure regions. Increases efficiency.



VORTEX GENERATORS

Creates vortex to retain airflow adhesion. Found ineffective for drag reduction.

DESIGN	SIZE	DRAG [N]	OFFSET
1	Large	0.021	+0.11
2	Medium	0.022	+0.04
3	Small	0.015	+0.05

EVALUATION

Increase in drag. Decided against implementation.

MULTI-ELEMENT WINGS

Classified front wing + additional element underneath found optimal.

SIDE PODS

FLAT SIDE PODS

Extensively tested double element.



TAPERED SIDE PODS

Redirect airflow around rear wheel.



DESIGN	DRAG	LIFT
Flat	0.2111	-0.1
Indented	0.2178	-0.1
Tapered	0.2149	-0.14

EVALUATION

Tapered effective but increased total drag. Flat chosen for manufacturability.

SIDE POD PLATES

Physical barrier to prevent rollover.



SIDE POD BARGEBOARD

Turning vane implemented. Reduced drag by 0.6%.



CENTRE OF MASS

Cutting away pockets to align CoM.

FRONT	REAR	VERT	HORIZ
None	None	7.1	71.6
2	8	4.74	72.0
8	8	4.95	71.6

EVALUATION

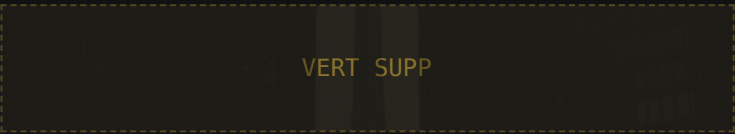
Final side pod design chosen with 8mm wall.

REAR WING

HORIZONTAL SUPPORTS



VERTICAL SUPPORTS



Vertical provided stability but too much drag.

TWISTED REAR WING

Angled to deflect towards low pressure zone.



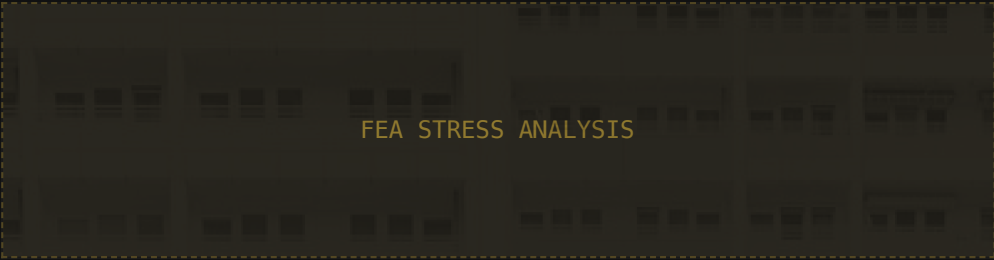
EVALUATION

Drag: 0.2542 [N]
Lift: -0.1896 [N]
Race Time: 1.063 [s]

Nosecone: Maintained innovation. Gentle coercion of airflow.

Front Wing: Redirected airflow to top of side pods.

Side Pods: Accommodated redirection.



FEA TESTING

Ran full car FEA at 10N. Max stress 195 MPa at tether guides (Titanium). Safe.

WIND TUNNEL TESTING

Difference of 0.02N in drag between CFD and Tunnel. Smoke visualization confirmed reattachment.



WEIGHT DISTRIBUTION

Pockets used to shift CoM forward between wheels. 2 deg canister angle.

PART INTEGRATION

Minimised individual components. Wheel support structures act as tether guide supports.

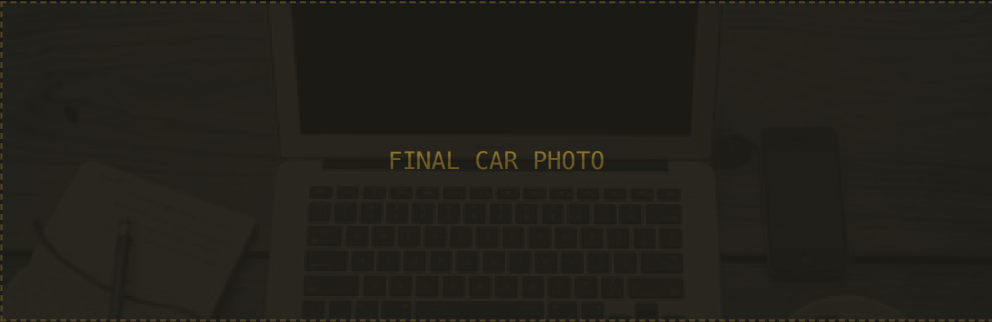


FINAL STATISTICS

Objectives: Decrease in drag of 17.95%.

Aerodynamics: 0.2542N Drag.

Mass: Kept as low as possible.



SOFTWARE CHOICE

CATIA V5

Main software. Precise surface control. Tree navigation.

AUTODESK FUSION 360

Used for rendering and FEA.

CAD ORGANISATION

MASTER SKETCHES

Control dimensions globally.

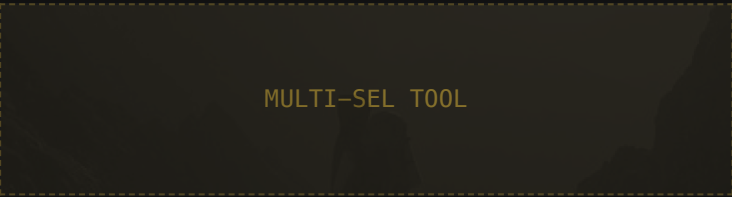


SURFACE MODELLING

Generative Shape Design used for complex aero surfaces.

MULTI-SELECTION SURFACE

Guides and profiles to create smooth body.



FILL

Used for nosecone of prototype 2.

SURFACE CONTINUITY

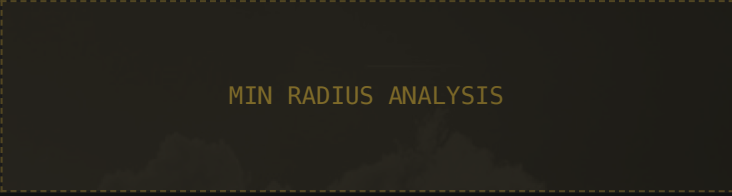
G0: Meeting. **G1:** Tangent (Filletts). **G2:** Curvature (Smoothest).

G1 identified as minimum for aero.

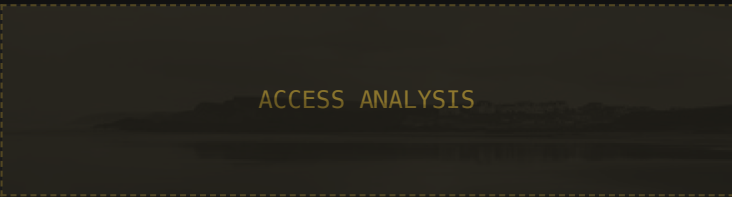


MFG PREPARATION

Account for CNC limits (3mm ball nose).



Accessibility Analysis in Fusion 360 to check undercuts.



OFFSET SPLIT SURFACES

0.1mm gap for glue.



MACHINING GOALS

- » Accurately produce 2 cars.
- » Compliant cars.
- » Mass close to 50g.

CNC EQUIPMENT

Denford 6600 Pro. 4-axis. 24000 RPM.



MACHINING TOLERANCE

IT12. 0.1mm clearance. Exclusion zones 1mm.

Milling plans optimized to minimize heat and tolerance creep.



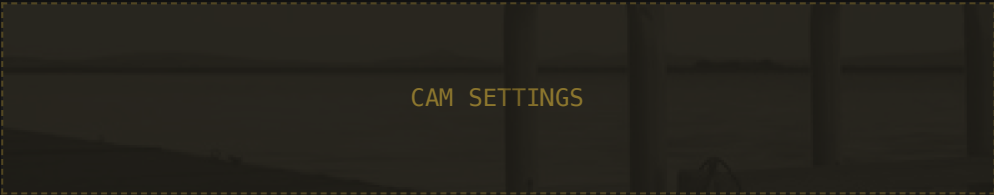
CAM

CAM SOFTWARE

Quick Cam Pro. Native integration.

SETTINGS

3000 mm/s feed. 45 deg raster.



QUALITY CONTROL

COMPLIANCE

Two person checking system. Scrutineering sessions performed.

GENERAL

Manufactured 6 cars. Outliers rejected.



Wheel bearings burnished and ranked.

PROCESS

SANDING & SEALING

360 to 400 grit. PVA/Water seal.

ASSEMBLY

Dry run. 5 min epoxy. Alignment jig.



PUTTYING

Wood putty for gaps.

PRIMER & FINISH

3 coats primer. 1-2 coats auto paint.



ASSEMBLY JIG

Negative impression for alignment. FDM PLA.



OUTSOURCING

NYLON PARTS

SLS 3D Printing. High yield strength.

PEEK WHEELS

CNC Turned. +/- 0.05mm.



WORKPLACE SAFETY

RISK	CAUSE	PROB	IMP	SCORE	CONTROL
Irritation	Dust	0.6	0.4	0.24	Extraction
Drowsiness	Fumes	0.4	0.6	0.24	Ventilation
Impact	Ejection	0.1	0.8	0.08	Guards
Cuts	Tools	0.1	0.2	0.02	Training

Specific information from SDS used.